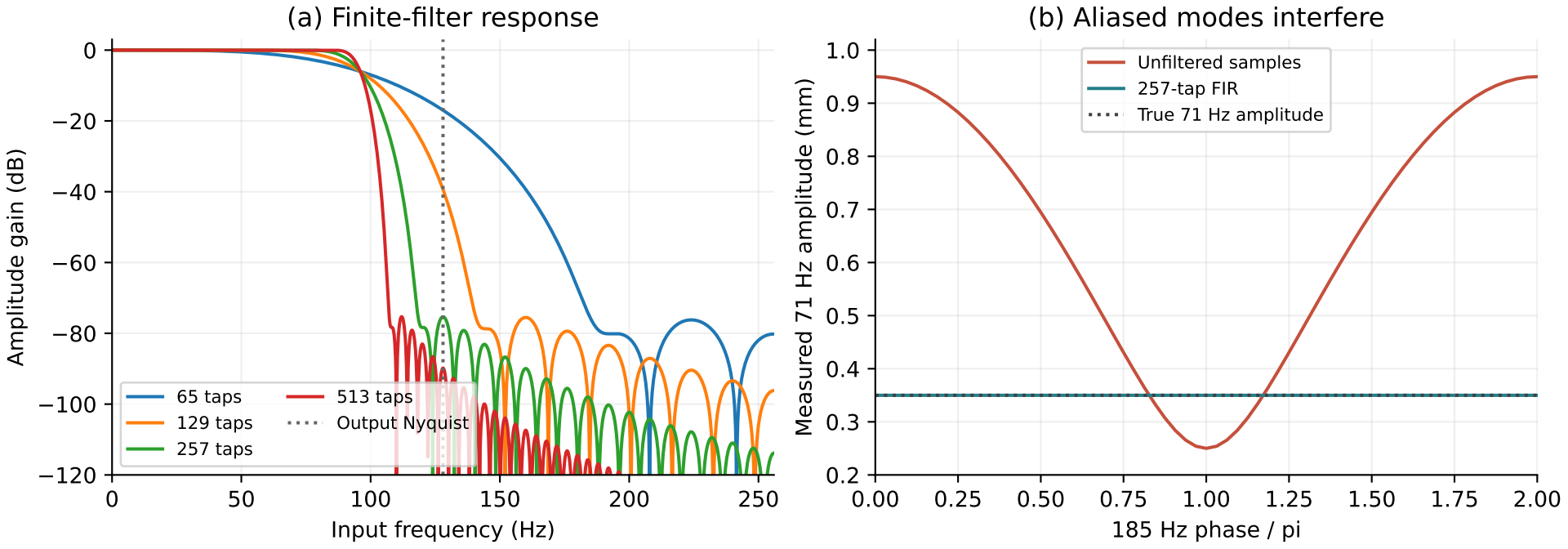
(a) Finite-filter response
Amplitude gain (dB)
0
−20
−40
−60
−80
−100
−120
0
50
100
150
200
250
Input frequency (Hz)
65 taps
129 taps
257 taps
513 taps
Output Nyquist
(b) Aliased modes interfere
Measured 71 Hz amplitude (mm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
185 Hz phase / pi
Unfiltered samples
257-tap FIR
True 71 Hz amplitude